BioMatch – Startup Input, Readiness & Matching Spec

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Version 2 of the BioMatch spec defines what a biotech startup enters, which questions it answers, how the Readiness Score (listing threshold 60/100) and the reciprocal Matching Score are computed, and how the platform stays NDA-free. Scope: therapeutics.

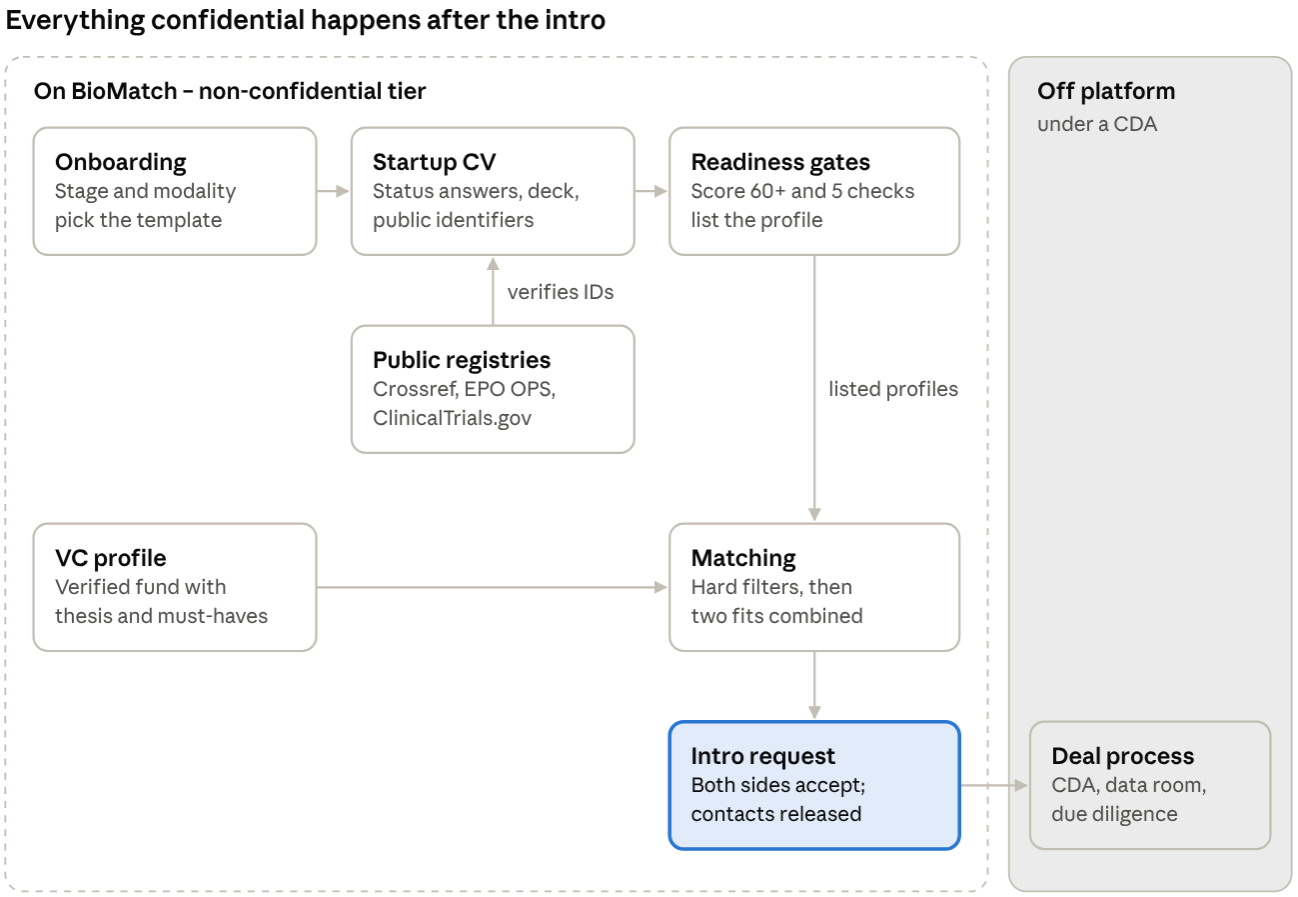
Full question bank: Question bank

Design principles

BioMatch is the non-confidential first tier of a biotech deal: it gets the right startup and the right VC into a first call, and everything after that call happens under a CDA, off the platform.

- **Non-confidential by design.** Pharma business development already works in tiers: a non-confidential deck and one-pager open the conversation, and the confidential package opens only after a CDA (Peony out-licensing playbook). BioMatch is that first tier. A non-confidential presentation explains what a technology does, not its inventive step, and covers unmet need, people, IP status, maturity, funding and the next research plan (Aarhus University guide) – the backbone of the Startup CV.
- **Structured over free text.** Status choices are scorable, comparable across startups, and leak far less than prose.
- **Stage-fair.** Each startup is scored against the template of its development band, so a discovery-stage spin-out is never penalised for data it cannot have yet.
- **Verify, don't trust.** High-value claims – a publication, a granted patent, a registered trial, an awarded grant – need a public identifier, which the platform checks against public registries.
- **Explainable.** Every score shows its drivers and the three actions that would raise it most.
- **Reciprocal.** A match surfaces only when both sides' must-haves pass, and the score weighs both sides' fit equally.

How it works



BioMatch flow · 8 steps, 2 zones

Startups and VCs meet on non-confidential, partly registry-verified data. Contacts are released only when both sides accept an intro; the CDA, data room and diligence happen off the platform.

Stage model

The readiness template follows the **development band** – what evidence can exist – while the **financing round** drives matching and a plausibility check. Bands map to the biomedical TRL scale used by European funders (DLR TRL scale for life sciences).

Band	TRL	Evidence that typically exists	Typical round	Template emphasis
D1 Discovery	1–3	Research idea and protocols; hypothesis testing and initial proof of concept in a limited number of in-vitro / in-vivo models	Pre-seed, seed, grants	Team, science, unmet need
D2 Lead and candidate	4	Proof of concept and early safety of a candidate in a defined lab or animal model; candidate nominated	Seed, Series A	Science, product plan, IP
D3 IND/CTA-enabling	5	Preclinical studies; IND or IMPD in preparation or submitted	Series A	IP and FTO, regulatory, CMC
D4 Clinical	6–7	IND/IMPD approved; Phase 1 executed; Phase 2 executed or completed	Series B, C, crossover	Human data, regulatory, CMC, market

Template = band × modality. A D2 small-molecule company answers the core D2 questions plus the small-molecule module; a D2 cell-therapy company answers the same core plus the cell-therapy module. Six modules exist: small molecule, biologics, cell therapy, gene therapy and editing, RNA and oligonucleotides, platform.

Plausibility check. A round outside the typical range for the band – for example a Series B at D1 – raises a consistency flag that the founder must confirm or correct.

Startup CV

The Startup CV has four parts: 18 profile fields (unscored, used for listing and matching), 46 scored core questions in nine dimensions, six modality modules with 19 scored questions, and a non-confidential pitch deck. A D2 small-molecule company answers 47 scored questions – about 45 minutes. Every question, answer option and point value is listed in the Question bank tab.

Profile fields (unscored)

ID	Field	Type	Used for
P01	Company name, website, logo	text, url, image	Listing
P02	One-line pitch (max 140 characters)	text	Search, cards, thesis similarity
P03	HQ country and city	dropdown	Geography fit
P04	Founding year	number	Display
P05	Origin + institution: university or institute spin-out, pharma carve-out, VC-created, independent	dropdown, text	Display, conflict check
P06	Modalities	multi-select	Modality module, modality fit
P07	Therapeutic areas	multi-select	Area fit
P08	Lead indication	text + category	Market questions, conflict check
P09	Company type: asset-centric, platform with lead asset, platform only	dropdown	Asset/platform fit
P10	Development stage (TRL)	dropdown	Selects the band template
P11	Current round	dropdown	Round fit, plausibility check
P12	Target raise (range) and amount committed	range, number	Ticket fit
P13	Lead investor: need a lead, have a lead, follow-on only	dropdown	Role fit
P14	Investor types sought	multi-select	Investor-type filter
P15	Preferred investor regions	multi-select	Geography fit
P16	Value-add wanted	multi-select	Value-add fit
P17	Target closing date	month	Timing fit
P18	Non-confidential pitch deck + attestation	PDF, checkbox	Listing gate, VC review

Scored core questions

Dimension	Questions	What they cover
Team	7	Full-time founders, drug-development experience, prior exits, who leads, hiring plan, advisory board, board
Science and evidence	8	Target validation, evidence package against band expectations, reproducibility, exposure at the site of action, safety liabilities, patient selection, positioning vs standard of care, 12-month kill/go experiment
Product and plan	5	Lead indication focus, Target Product Profile, pipeline depth, milestone plan, months to the next value-inflection point
IP and exclusivity	5	Filing status, ownership and licence, freedom to operate, years of protection after launch, other exclusivity
Regulatory and clinical	5	Pathway, agency interaction, expedited designations, first-in-human plan, clinical status with registry IDs
CMC	4	Manufacturing route, GMP material, cost of goods, supply risks
Market	5	Sourced patient population, standard of care, competitive map, partnering thesis with comparable deals, pricing hypothesis
Financing	3	Use of funds by milestone, runway, share of the round committed
External validation	4	Publications, grants, pharma collaborations, accelerators

The science questions follow AstraZeneca's 5R framework – right target, tissue, safety, patient and commercial potential. AstraZeneca reports that applying it lifted success from candidate nomination to Phase 3 completion from 4% (2005–2010) to 19% (2012–2016) (Morgan et al., Nat Rev Drug Discov 2018).

Modality modules

Module	Scored questions	What investors probe
Small molecule	4	Chemical matter stage, ADME/DMPK, selectivity, synthesis scale-up
Biologics	3 + format	Developability, cell line, immunogenicity
Cell therapy	3 + cell source	Process status, potency assay, vein-to-vein time and COGS
Gene therapy and editing	3 + delivery	Vector supply, pre-existing immunity and redosing, off-target or integration safety
RNA and oligonucleotides	3	Delivery to the target tissue, duration of effect, tolerability chemistry
Platform	3 + business model	Programs generated, external validation, lead asset nominated

CMC carries extra weight in the cell and gene therapy modules: in one study, more than 20% of clinical holds on cell and gene therapy trials also involved CMC issues (BioProcess Online).

Pitch deck rules

- PDF only, max 25 MB, ideally no more than 15 main slides; the title slide says "Non-confidential". BIO's licensing guidance gives the same advice and adds: no P&L, cash flow or other financial statements (BIO, Cupit and Sinatra).
- The builder shows a slide checklist (not scored): problem, solution (what it does, not how), evidence summary, pipeline and milestones, IP status, regulatory path, team, market and competition, comparable deals, ask and use of funds. A comparable-deals slide matters because partnering or acquisition is the most likely outcome for a therapeutics startup, and early decks often skip the regulatory slide (Pillar VC).
- Every upload passes the confidentiality scan (see Verification and privacy).
- The platform renders a forwardable one-page summary from the CV, so VCs can share a startup internally without the deck.

Never collected

- Chemical structures, sequences, constructs, formulations or process parameters
- Unpublished raw data or figures that are not in the public deck

- Content of unpublished patent applications; FTO opinions
- Term sheets, cap tables, valuations, financial models, contracts
- Patient-level data of any kind

Biotech VC criteria

Biotech investors judge the science first, then its reproducibility, then the team: Atlas Venture's Bruce Booth names transformative science, confidence that it replicates, and experienced people as the filters, and about half of Atlas's investments start with a seed phase that only validates the science (McKinsey interview). Interviews with biotech VCs rank management team, market size, IP, business model and product status as the top five factors, with the emphasis moving from team and IP at start-up stage to product and clinical results later (NYCU study).

Criteria library

Each criterion maps to the question that scores it and to the expectation per band: **M** must, **E** expected, **B** bonus, **–** not yet relevant.

Area	Criterion	Good answer	Question	D1	D2	D3	D4
Team	Commitment	Full-time CEO and scientific lead	TM01	M	M	M	M
Team	Drug-development judgment	Someone who has taken an asset into the clinic	TM02	E	M	M	M
Team	Track record	Prior VC-backed company or exit	TM03	B	E	E	E
Team	Leadership	Experienced biotech CEO in place or being recruited	TM04	B	E	M	M
Team	Advisors and board	Advisory board with KOLs; independent director by D3	TM06, TM07	B	E	E	M
Science	Transformative potential	Serious unmet need; first- or best-in-class	SC07, MK02	M	M	M	M
Science	Target confidence	Human genetic or clinical evidence for the target	SC01	E	E	M	M
Science	Reproducibility	Key result repeated by a CRO, second lab or partner	SC03	B	M	M	M
Science	Evidence package	All evidence expected for the band	SC02	M	M	M	M

Area	Criterion	Good answer	Question	D1	D2	D3	D4
Science	Exposure and safety	Exposure at the site of action; safety screen done	SC04, SC05	B	E	M	M
Science	Patient selection	Biomarker or selection strategy	SC06	B	B	E	M
Science	De-risking plan	Funded kill/go experiment in the next 12 months	SC08	E	M	M	E
Product	Focus	One lead indication, follow-ons mapped	PD01	E	M	M	M
Product	Target Product Profile	Written TPP reviewed with clinicians	PD02	B	E	M	M
Product	Pipeline depth	Backup candidate or second program	PD03	–	B	E	E
Product	Capital efficiency	Milestones with dates and budget for this round	PD04, PD05, FN01	M	M	M	M
IP	Ownership	Company-owned or exclusive licence signed	IP02	E	M	M	M
IP	Filing	Priority filing at seed, PCT by Series A, issued claims by Series B	IP01	B	E	M	M
IP	Freedom to operate	Professional FTO analysis	IP03	–	B	M	M
IP	Exclusivity horizon	Patent term plus regulatory exclusivity well past launch	IP04, IP05	–	B	E	M
Regulatory	Pathway	Pathway and key requirements mapped	RC01	B	E	M	M
Regulatory	Agency interaction	Pre-IND or scientific advice held	RC02	–	B	M	M
Clinical	Realistic plan	First-in-human outline with population, endpoints, sites	RC04	–	B	M	M
Clinical	Human data	Registered trials with safety and early efficacy	RC05	–	–	–	M
CMC	Manufacturing route	CDMO or own GMP capability	CM01	–	E	M	M
CMC	GMP readiness	GMP material for toxicology and Phase 1	CM02	–	–	M	M
Market	Sized opportunity	Sourced patient numbers and price analogues	MK01	E	M	M	M
Market	Competitive position	Competing programs mapped by mechanism and stage	MK03	E	M	M	M

Area	Criterion	Good answer	Question	D1	D2	D3	D4
Market	Exit logic	Partnering or M&A thesis with comparable deals	MK04	B	E	M	M
Financing	Runway	18+ months after close, milestones funded	FN01, FN02	M	M	M	M
Validation	Third-party signals	Peer-reviewed paper, grant, pharma collaboration	EV01–EV04	B	E	E	E

The IP rows follow a stage-by-stage diligence framework: chain of title matters most at seed, freedom to operate is often the most pressing question at Series A, and by Series B investors expect issued claims with meaningful scope (Sterne Kessler via JD Supra). Target confidence carries weight because AstraZeneca's pipeline review found projects with stronger target validation, especially a genetic link, less likely to fail (BioPharma Dive).

Red flags and how the platform checks them

Investors walk away early over unclear IP ownership, an inexperienced team or unrealistic clinical timelines (Excedr). The platform turns each into a check, so founders fix it before a VC sees it.

Red flag	Platform check	Effect
Unclear IP ownership	IP02 = "university-owned, no agreement" at D3 or D4	Critical: IP dimension capped at 50%, shown to VCs as an open question
Unrealistic timelines	Months to next inflection or first-in-human date far below band norms	Warning: founder must confirm
No independent validation	SC03 = "single lab" at D2 or later	Open question shown to VCs
Inexperienced team, no advisors	TM02 = "none" and TM06 = "none" at D2 or later	Open question shown to VCs
No regulatory plan	RC01 = "not defined" at D2 or later	Warning
Short runway	FN02 = under 12 months	Warning
No focus	PD01 = "undecided" at D3 or D4	Warning

VC profile

The VC profile mirrors the Startup CV field by field, adds fund capacity, and marks each preference as **must-have** (hard filter) or **nice-to-have** (weighted). Only verified VCs – work email on the fund's domain, LinkedIn check, manual review – can see startup profiles.

Field	Type	Matches	Used as
Fund name, website, logo, partners (name, focus, LinkedIn)	text, rows	–	Display
Investor type: specialist biotech VC, generalist VC with life-science team, corporate VC, family office, angel network, impact or public fund	dropdown	P14	Startup's filter
Accepts inbound startups (venture-creation firms may not)	yes / no	–	Hard
Investment rounds	multi-select	P11	Hard
Development bands accepted (D1–D4)	multi-select	P10	Must or nice
Modalities	multi-select	P06	Must or nice
Therapeutic areas	multi-select	P07	Must or nice
Asset vs. platform preference	dropdown	P09	Weighted
Initial ticket (min–max) and typical total per company	range	P12	Hard
Role: lead, co-lead, follow	multi-select	P13	Hard
Company HQ regions	multi-select	P03, P15	Must or nice
Minimum IP position	dropdown	IP01, IP02	Must or nice
Requires human data	yes / no	RC05	Hard
Minimum Readiness Score (default 60) and minimum verified share	number	Readiness	Hard

Field	Type	Matches	Used as
Fund vintage, end of investment period, reserve ratio	year, %,	–	Fund capacity
Value-add: operating partners, company building, clinical and regulatory team, CMC network, pharma BD network, recruiting, follow-on syndicate	multi-select	P16	Weighted
Board seat expectation	dropdown	–	Display
Decision speed: weeks from first call to term sheet	number	P17	Weighted
Portfolio companies (public) with target class and lead indication	list	P08	Conflict check
Thesis statement (max 500 characters)	text	P02, SC07	Thesis similarity
Exclusions (e.g. no devices, no single-asset companies)	multi-select	P06, P09	Hard

Why fund capacity matters. A fund can usually make new investments only during its investment period – typically three to five years from its close – and afterwards writes follow-on cheques only (VC Beast). A profile past that window is hidden from new startups automatically.

Corporate VCs answer one extra question – which parent-company strategic areas they cover – because a CVC looks for strategic fit as well as financial return (Bird & Bird).

Readiness Score

A startup is listed when it passes five gates and scores at least **60/100** against its band template. The score is a weighted sum of nine dimension scores, and each dimension is built from answer points between 0 and 1.

Weights per band

Each column sums to 100. Weight moves from team and science towards IP, regulatory and CMC as a company matures.

Dimension	D1	D2	D3	D4
Team	25	20	15	15

Dimension	D1	D2	D3	D4
Science and evidence	30	25	20	20
Product and plan	10	15	15	10
IP and exclusivity	10	10	15	10
Regulatory and clinical	5	5	10	15
CMC	0	5	10	10
Market	10	10	5	10
Financing	5	5	5	5
External validation	5	5	5	5

Cell and gene therapy modifier: if the cell-therapy or gene-therapy module is active, 5 points move from Science to CMC at every band.

How a dimension is scored

$$\text{Readiness} = \sum_{d} w_d(\text{band}) \cdot s_d \qquad s_d = \min\left(1,\; \overline{R}_d + 0.1 \cdot \overline{O}_d\right)$$

- $\overline{R}$ is the mean of the required answers in dimension d, $\overline{O}$ the mean of its optional answers (unanswered = 0). Optional answers add up to 10% – a bonus, never a penalty.
- A dimension with no required question at a band takes its best optional answer. External validation always works this way: one strong signal is enough.
- The evidence package (SC02) is scored against the evidence expected at the band; each extra item adds 0.05.
- Positioning (SC07) needs three differentiation points; without them it scores 70%.

Proof rule and evidence confidence

High-value answers count in full only with a public identifier: a granted patent needs its publication number, a paper or preprint its DOI, an awarded grant its grant ID, human data or clinical status a trial registry ID, and designations, comparable deals or paid partnerships a public link. Without it, the answer scores like the best lower option.

Evidence confidence (0–1) shows VCs how much of a profile is backed: registry-verified identifiers count 1, links count 0.5, and a profile without proof-eligible claims is neutral at 0.5. It appears as a badge and feeds the Matching Score.

Gates

1. Non-confidential deck uploaded, attestation ticked, scan findings resolved.
2. At least one full-time founder or executive.
3. All profile fields and required questions for the band and modules answered.
4. No unconfirmed warnings.
5. Profile updated within the last 6 months.

Consistency checks

ID	Trigger	Severity
F1	Series A at D1, or Series B or later at D1–D2	Warning
F2	D4 without a verified trial registry ID	Warning
F3	University-owned IP without agreement at D3–D4	Critical
F4	Human data reported below D4	Warning
F5	Next value-inflection point under 6 months away	Warning
F6	Runway under 12 months after close	Warning
F7	No lead indication at D3–D4	Warning
F8	Key result not reproduced, D2 or later	Open question
F9	No drug-development experience on team or advisory board, D2 or later	Open question
F10	No regulatory pathway, D2 or later	Warning

A **warning** blocks listing until the founder confirms or corrects it. An **open question** is shown to VCs next to the profile. A **critical** flag caps its dimension at 50% and is shown to VCs.

Levels and feedback

Score	Level	Unlocks
Under 60, or a gate failed	Draft	Visible only to the startup
60–74	Listed	Visible to verified VCs; can browse and contact VCs

Score	Level	Unlocks
75–89	Investor-ready	Badge; ranked higher in VC search
90–100	Top profile	Featured in the weekly VC digest

The builder re-scores every possible next step and shows the three with the largest gain. **Quick wins** close an open question or add a missing identifier; **milestones** mark real progress, such as a completed FTO opinion.

Worked example

Nuvaris Therapeutics (fictional) is a Berlin seed-stage company with a small-molecule candidate for glioblastoma – band D2 plus the small-molecule module, 47 scored questions. It scores **65.2** and is listed.

Dimension	Weight	Dimension score	Points
Team	20	0.70	13.9
Science and evidence	25	0.66	16.6
Product and plan	15	0.74	11.1
IP and exclusivity	10	0.51	5.1
Regulatory and clinical	5	0.50	2.5
CMC	5	0.62	3.1
Market	10	0.70	7.0
Financing	5	0.58	2.9
External validation	5	0.60	3.0
Total	**100**		**65.2**

Earlier in the builder the same company scored 61.3 but stayed a draft: two required answers were open (TPP and partnering thesis) and the epidemiology source link was missing. The builder's top three steps then were: answer the TPP question (up to +3.0), answer the partnering thesis (up to +2.5), and map the regulatory pathway (+2.5). All numbers come from the reference implementation delivered with this spec.

Matching Score

The Matching Score (0–100) is the geometric mean of two fits – how well the startup fits the VC's thesis, and how well the VC fits the startup's needs – so a pair only scores high when both sides would say yes. This is the reciprocal-recommender principle used by dating and recruiting platforms, where a match counts only if both sides accept and each side's preferences are fused into one mutual score (Palomares et al. 2020).

Step 1 – Hard filters

Any failed filter hides the pair; the reason is logged for the side that set it.

- Round outside the VC's rounds, or the VC's minimum cheque above the whole round
- Startup needs a lead, and the VC does not lead
- VC must-haves not met: band, modality, therapeutic area, geography, minimum IP position, human data
- Readiness below the VC's minimum (default 60)
- VC type not among the investor types the startup wants
- Fund past its investment period, fund not accepting inbound startups, or company type excluded

Step 2 – Two fits

Side	Component	Weight	Scoring rule (0–1)
VC fit	Therapeutic area	20	Share of the startup's areas the VC covers
VC fit	Modality	15	Share of the startup's modalities the VC covers
VC fit	Development band	15	Accepted 1.0 · neighbouring band 0.5 · else 0
VC fit	Round	15	Accepted 1.0 · neighbouring round 0.5 · else 0
VC fit	Geography	10	Same country 1.0 · same region 0.7 · global mandate 0.6 · else 0
VC fit	Asset vs. platform	5	Match 1.0 · VC neutral 0.7 · mismatch 0
VC fit	Thesis similarity	10	Embedding similarity of VC thesis vs. pitch and differentiation points (demo: keyword overlap)
VC fit	Readiness and evidence	10	0.7 × (Readiness − 60) / 40 + 0.3 × evidence confidence

Side	Component	Weight	Scoring rule (0–1)
Startup fit	Ticket and role	30	Takes the role needed and max cheque ≥ 20% of the round 1.0 · ≥ 10% 0.6 · else 0.3
Startup fit	Value-add	25	Share of the startup's wanted value-add the VC offers · nothing wanted 0.7
Startup fit	Investor type	15	Wanted type 1.0 · startup open to all 0.7
Startup fit	Fund capacity	20	In investment period with ≥ 40% reserves 1.0 · lower reserves 0.6
Startup fit	Decision speed	10	Typical weeks to term sheet fit before target close 1.0 · else 0.4

Step 3 – Combine and explain

```
\text{Match} = \sqrt{F_{\text{VC}} \cdot F_{\text{startup}}} \; - \; 10 \cdot \mathbb{1}_{\text{conflict}}
```

The geometric mean lets the weaker side pull the pair down: fits of 90 and 40 give 60, not the 65 an average would. A conflict is a VC portfolio company with the same target and lead indication. Every match card names its three strongest components and its biggest gap.

Examples for Nuvaris Therapeutics

All four funds are fictional; numbers come from the reference implementation.

VC	VC fit	Startup fit	Match	Why
Rheinland Bio Ventures	89.6	100	95 – Strong	Covers oncology and small molecules, leads DACH seed rounds, offers the company-building, CMC and pharma-BD help Nuvaris wants. Gap: readiness only 65.
Helvetia Life Science Fund	79.9	75.3	68 – Good	Area, modality and ticket fit; −10 for a portfolio company with the same target and indication. Gap: value-add.

VC	VC fit	Startup fit	Match	Why
Northstar Generalist Ventures	31.4	67.0	46 – Weak	Round and ticket fit, but no area or modality overlap. Hidden by default.
Atlantic Precision Partners	–	–	Filtered out	Seed is outside its rounds, small molecules are outside its must-have modalities, and readiness is below its minimum of 70.

Display and contact

Score	Label	Behaviour
75–100	Strong match	Top of search; in both sides' weekly digest
50–74	Good match	Shown in search
Under 50	Weak match	Hidden by default, visible with a filter

Either side sends an intro request (max 500 characters, no attachments); a startup can have 10 open requests at a time. The other side accepts or passes with a structured reason – too early, outside thesis, team gap, IP concern, round size, timing – and pass reasons flow back into the startup's builder as feedback. On acceptance both sides get each other's contact email, and the platform steps out: CDA, data room and diligence happen elsewhere.

Verification and privacy

Three layers keep the platform credible and NDA-free: registry checks back up claims, a scanner stops leaks before they are stored, and access rules limit who sees what.

Registry checks

Claim	Identifier	Automatic check
Paper or preprint	DOI	Crossref REST API `/works/{doi}` – public, no sign-up (Crossref)

Claim	Identifier	Automatic check
Published or granted patent	Publication number	EPO Open Patent Services `/published-data/publication/epodoc/{number}/biblio` – free OAuth registration (OPS endpoint reference)
Clinical trial	NCT ID	ClinicalTrials.gov API v2 `/api/v2/studies/{NCT_ID}` – no API key, about 50 requests per minute (API reference)
NIH grant	Project number	NIH RePORTER API v2 at api.reporter.nih.gov (CASRAI guide)
EU or national grant, designation, partnership, comparable deal	Public link	Link reachable and spot-checked by the team

A check confirms that the identifier exists and that the company or a named founder appears in the record. Verified identifiers raise evidence confidence; failed checks are shown to the founder, never silently dropped.

Upload and text scanning

- **Keyword scan** of decks and free text for "confidential", "proprietary", "do not distribute", "under CDA" or "trade secret": the founder must confirm the content is public or replace the file.
- **Pattern scan** that blocks sequence-like and structure-like strings: runs of 20 or more nucleotide letters, 30 or more amino-acid letters, SMILES-like strings and "SEQ ID NO".
- **Metadata stripping** before storage: author names, comments, tracked changes and speaker notes.
- Decks open in a viewer watermarked with the viewer's name and fund; download is the founder's choice, and the startup sees a view log.

Access rules

- Startup profiles are visible only to verified VCs; VC profiles are visible to listed startups.
- Intro requests are text only, with no attachments; contact details are released only after both sides accept.
- The terms state that the platform holds no confidential information, is not party to any CDA or NDA, and gives no investment advice; founders attest that every upload is non-confidential.

- Data is hosted in the EU and deleted on request; profiles inactive for 24 months are archived.

Mockup

The clickable mockup of the startup input lives on a design canvas: BioMatch – Startup CV builder. It shows the fictional Nuvaris Therapeutics on the D2 + small-molecule template and runs the same scoring logic as the reference implementation.

Screen	What it shows	Interactive
1 Stage and template	TRL, modality, company type and round pick the template; question count, time, top weights and what investors expect update live; unusual round warns	Yes
2 Startup CV builder	Sections by dimension, answer chips with points, evidence checklist against band expectations, proof fields with registry status, live Readiness ring with the listing line, next best steps, listing checks, dimension bars, evidence confidence	Yes – the score recalculates on every click
3 Pitch deck	Upload, confidentiality scan (blocked sequence, "proprietary" check, stripped metadata), attestation, slide checklist, viewer and download controls	Yes
4 What investors see	Profile card, key facts, differentiation, evidence, documents, readiness and evidence badges, best-matching funds with reasons	Static

Try this in screen 2: the builder opens at 61.3 as a draft with two required answers open. Choosing “Draft” for the TPP and “Thesis stated” for the partnering thesis, then adding the epidemiology source link, lists the profile at 65.1.

Hackathon build plan

The engine is ready: question-bank.json holds every question and point value, and scoring.js computes readiness, actions and matches in plain JavaScript with no dependencies (browser or Node). The team builds the screens and the data around it.

Why now, for the pitch: in the first half of 2026 biotech Series A rounds fell to a ten-year low of 87 while the median round rose to a decade-high $29.1M (BioCentury). Fewer, larger bets make precise matching worth more to both sides.

Build order

- [ ] Render the Startup CV from question-bank.json (template = band × modules) and save answers
- [ ] Wire scoring.js into the builder: live score, gates, checks, next best steps
- [ ] Onboarding screen: stage, modality, company type, round
- [ ] Deck upload: PDF text extraction, keyword and sequence scan, attestation
- [ ] Registry checks that need no key: Crossref DOI and ClinicalTrials.gov NCT; stub EPO OPS until OAuth is set up
- [ ] VC profile form and five seeded fictional funds
- [ ] Match lists for both sides, intro request with accept or pass reasons
- [ ] Seed 8–10 fictional startups across all four bands and several modalities

Three-minute demo script

1. **Problem (0:00).** Biotech founders cannot share data before an NDA, and VCs read decks that miss their thesis.
2. **Onboarding (0:30).** Pick TRL 4 and small molecule: template D2, 47 questions, about 45 minutes.
3. **Builder (1:00).** The score sits at 61.3 as a draft. Answer the two open questions and add a source link: listed at 65.1. The link shows the proof rule at work: without it, that answer scored like the level below.
4. **Deck scan (1:45).** The scanner blocks a sequence on slide 11; confirm the "proprietary" mention on slide 7; the deck clears.
5. **Matches (2:15).** Rheinland 95 (strong), Helvetia 68 (portfolio conflict), Atlantic filtered out by its own must-haves.
6. **Close (2:45).** The platform never holds confidential data – the CDA starts after the intro.

Sources

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- Unleashing the power of entrepreneurship in science – McKinsey interview with Bruce Booth, 2016
- A Study of Investment Evaluation on Biotechnology Venture Capital – Chou and Yang, NYCU, 2007
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- Reciprocal Recommender Systems – Palomares et al., 2020
- Crossref REST API – Crossref
- EPO OPS endpoint reference – epo-ops-sdk
- ClinicalTrials.gov API v2 reference
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- Series A rounds grew larger as deal count fell in 1H26 – BioCentury, 2026

Question bank

Every scored question of the Startup CV with its answer options and points (0–1). Columns D1–D4: **R** required, **O** optional bonus, **–** hidden at that band. An identifier in brackets is needed for the answer to count in full (proof rule); without it the answer scores like the best lower option. The same bank ships as question-bank.json with the reference scoring engine.

Core questions

Team (7)

ID	Question	Answer options and points	D1	D2	D3	D4
TM01	How many founders or executives work full-time for the company?	None yet **0** · One **0.6** · Two or more **1**	R	R	R	R
TM02	What is the deepest drug-development experience in your leadership team?	None **0** · Academic translational research only **0.3** · Advisor with industry R&D experience **0.5** · Executive with industry R&D experience **0.8** · Executive who has taken an asset into the clinic **1**	R	R	R	R
TM03	Has anyone in the founding team built a company before?	No **0** · Yes, a VC-backed company **0.6** · Yes, with an exit or licensing deal **1** (public link)	O	R	R	R
TM04	Who leads the company today?	Scientific founder, first operating role **0.4** · Founder with a prior operating role **0.7** · Experienced biotech CEO **1**	O	R	R	R
TM05	Do you have a hiring plan for key gaps (e.g. CMO, CMC lead, CFO, BD)?	No plan yet **0** · Gaps named **0.5** · Gaps named with timing, funded by this round **1**	O	R	R	R
TM06	Who advises you scientifically and clinically?	No advisors yet **0** · Informal advisors **0.4** · Named advisory board with at least two KOLs in the indication **1** (public names)	O	R	R	R
TM07	How is your board composed?	Founders only **0.3** · Includes an investor director **0.7** · Includes an independent industry director **1**	–	O	R	R

Science and evidence (8)

ID	Question	Answer options and points	D1	D2	D3	D4
SC01	What is the strongest evidence that your target drives the disease?	Literature hypothesis **0.2** · Functional validation in vitro **0.4** · Validation in an in-vivo disease model **0.6** · Human genetic evidence **0.9** · Clinical validation of the mechanism **1**	R	R	R	R
SC02	Which evidence do you have today? Tick all that apply.	Expected per band – D1: in-vitro proof of concept, target engagement · D2: + in-vivo efficacy, PK/PD or dose-response · D3: + early (non-GLP) safety, GLP toxicology · D4: + human safety data (trial ID). Score = share of expected items + 0.05 per extra item, max 1. Human efficacy signal (trial ID) counts as an extra	R	R	R	R
SC03	Has your key result been reproduced?	Single lab, single series of experiments **0.2** · Reproduced in-house in independent experiments **0.5** · Reproduced by a CRO or a second academic lab **0.9** · Reproduced by a pharma partner **1**	O	R	R	R
SC04	Have you shown exposure at the site of action?	Not yet assessed **0** · Predicted by modelling **0.4** · Measured in animals **0.8** · Measured in humans **1**	O	R	R	R
SC05	How have you assessed safety liabilities?	Not yet assessed **0** · Literature and target-biology review **0.4** · Experimental screen (off-target panel or early tox) **0.8** · GLP toxicology complete **1**	O	R	R	R
SC06	How will you select the right patients?	No strategy yet **0** · Hypothesis **0.4** · Biomarker identified **0.7** · Biomarker assay or companion diagnostic plan **1**	O	O	R	R
SC07	How is your product positioned against the standard of care?	First-in-class **1** · Best-in-class with a stated advantage **0.9** · Repurposing with new IP **0.6** · Fast follower **0.5** – × 0.7 without three differentiation points	R	R	R	R

ID	Question	Answer options and points	D1	D2	D3	D4
SC08	Is your next kill/go experiment defined?	Not defined **0** · Defined **0.6** · Defined, funded and scheduled within 12 months **1**	R	R	R	R

Product and plan (5)

ID	Question	Answer options and points	D1	D2	D3	D4
PD01	How focused is your lead indication?	Several candidates, undecided **0.3** · Lead indication chosen **0.7** · Lead chosen, follow-on indications mapped **1**	R	R	R	R
PD02	Do you have a Target Product Profile (TPP)?	No **0** · Draft **0.5** · Written and reviewed with clinicians or advisors **1**	O	R	R	R
PD03	How deep is your pipeline?	Single program **0.5** · Lead plus backup candidate **0.8** · Lead, backup and a second program **1**	–	O	R	R
PD04	What is your plan to the next value-inflection point?	No plan yet **0** · Milestones listed **0.5** · Milestones with dates and budget, funded by this round **1**	R	R	R	R
PD05	How many months to your next value-inflection point?	Number. ⩽ 24 → 1 · 25–36 → 0.7 · > 36 → 0.4	R	R	R	R

IP and exclusivity (5)

ID	Question	Answer options and points	D1	D2	D3	D4
IP01	What is the most advanced IP filing for your lead program?	None yet **0** · Invention disclosed to the tech-transfer office **0.2** · Priority application filed **0.5** · PCT application filed **0.75** · Patent granted in a major market **1** (patent publication number)	R	R	R	R
IP02	Who owns the core IP?	University or institute, no agreement yet **0.2** · Option agreement signed **0.5** · Licence terms agreed **0.7** · Exclusive licence signed **0.9** · Company-owned **1**	R	R	R	R

ID	Question	Answer options and points	D1	D2	D3	D4
IP03	Has freedom to operate been assessed?	Not assessed **0** · Informal landscape search **0.4** · Professional FTO analysis in progress **0.7** · Professional FTO opinion completed **1**	–	O	R	R
IP04	How many years of protection remain after expected launch?	Unknown **0** · Under 5 years **0.3** · 5-10 years **0.7** · Over 10 years **1**	–	O	R	R
IP05	Which additional exclusivity applies? Tick all that apply.	Orphan exclusivity likely **0.7** · Regulatory data exclusivity **0.5** · Process know-how or trade secrets **0.5** – best ticked option counts	O	O	O	O

Regulatory and clinical (5)

ID	Question	Answer options and points	D1	D2	D3	D4
RC01	How defined is your regulatory pathway?	Not yet defined **0** · Target agencies named **0.5** · Pathway and key requirements mapped **1**	O	R	R	R
RC02	Have you interacted with a regulator?	No **0** · Meeting planned with a date **0.4** · INTERACT, pre-IND or scientific advice held **1**	–	O	R	R
RC03	Which status do your expedited designations have (orphan, fast track, breakthrough, RMAT, PRIME)?	None **0** · Planned **0.3** · Applied **0.5** · Granted **1** (public link)	O	O	O	O
RC04	How developed is your first-in-human plan?	Not started **0** · Outline: population and endpoints **0.5** · Protocol synopsis with CRO or sites identified **1**	–	O	R	R
RC05	What is your clinical status?	Phase 1 ongoing **0.5** · Phase 1 complete **0.7** · Phase 2 ongoing **0.8** · Phase 2 readout available **1** – every option needs a trial registry ID (NCT or EU CT number)	–	–	–	R

CMC (4)

ID	Question	Answer options and points	D1	D2	D3	D4
CM01	How will you manufacture?	Not defined **0** · Research-grade material only **0.3** · CDMO identified **0.6** · CDMO contracted or own GMP capability **1**	–	R	R	R
CM02	What is your GMP material status?	Not planned **0** · Planned with a date **0.5** · GMP batch produced **1**	–	–	R	R
CM03	Have you estimated cost of goods at commercial scale?	Not assessed **0** · Qualitative view **0.5** · COGS estimate **1**	–	O	O	R
CM04	Have you assessed critical supply risks?	Not assessed **0** · Identified **0.5** · Mitigated (second source or reserved capacity) **1**	–	O	O	O

Market (5)

ID	Question	Answer options and points	D1	D2	D3	D4
MK01	How have you sized the patient population for your lead indication?	Not sized **0** · Estimate without a source **0.4** · Sourced epidemiology **0.8** (public link) · Sourced epidemiology plus price analogues **1** (public link)	R	R	R	R
MK02	Have you described the current standard of care and its limitations?	No **0** · Yes **1**	R	R	R	R
MK03	How have you mapped the competition?	Not mapped **0** · Competitors named **0.5** · Competing programs mapped by mechanism and stage, with positioning **1**	R	R	R	R
MK04	What is your partnering or exit thesis?	None yet **0** · Thesis stated **0.5** · Thesis with at least two comparable deals **1** (public link)	O	R	R	R
MK05	Do you have a pricing and reimbursement hypothesis?	No **0** · Analogue-based hypothesis **0.6** · Payer input gathered **1**	–	–	O	R

Financing (3)

ID	Question	Answer options and points	D1	D2	D3	D4
FN01	How are the funds from this round allocated?	Not specified **0** · Budget categories **0.4** · Milestones with budget **1**	R	R	R	R
FN02	What runway will you have after this round closes?	Under 12 months **0.3** · 12-18 months **0.7** · Over 18 months **1**	R	R	R	R
FN03	How much of the round is already committed?	None **0.3** · Under 25% **0.6** · 25% or more, or a lead is secured **1**	O	O	O	O

External validation (4)

ID	Question	Answer options and points	D1	D2	D3	D4
EV01	Is your core science published?	No **0** · Preprint **0.6** (DOI) · Peer-reviewed paper **1** (DOI)	O	O	O	O
EV02	Have you received non-dilutive funding?	None **0** · Applied **0.3** · Awarded (e.g. EIC, GO-Bio, EXIST, NIH SBIR, BARDA) **1** (grant ID or award page)	O	O	O	O
EV03	Do you collaborate with industry?	No **0** · Research collaboration or MTA with pharma **0.7** · Paid partnership or option deal **1** (public link)	O	O	O	O
EV04	Have you been selected by an accelerator or won a competition?	No **0** · Yes **0.8** (public link)	O	O	O	O

Modality modules

A module switches on when its modality is selected (P06) or, for Platform, when the company type is a platform (P09). The Dimension column shows where each answer counts.

Small molecule (4 scored)

ID	Dimension	Question	Answer options and points	D1	D2	D3	D4
SM01	Product and plan	What stage is your chemical matter?	Hit series **0.3** · Lead series **0.6** · Development candidate nominated **1**	O	R	R	R

ID	Dimension	Question	Answer options and points	D1	D2	D3	D4
SM02	Science and evidence	What ADME / DMPK data do you have?	Not started **0** · In-vitro ADME **0.5** · In-vivo PK in two or more species **1**	O	R	R	R
SM03	Science and evidence	How have you profiled selectivity?	Not yet **0** · Focused panel **0.5** · Broad safety panel **1**	O	R	R	R
SM04	CMC	What is your synthesis scale-up status?	Not assessed **0** · Route identified **0.5** · Multi-gram or kilogram scale achieved **1**	–	O	R	R

Biologics (3 scored)

ID	Dimension	Question	Answer options and points	D1	D2	D3	D4
BI01	–	Which format is your biologic?	Monoclonal antibody · Bispecific · Antibody-drug conjugate · Fusion protein · Peptide · Other – not scored	R	R	R	R
BI02	CMC	Have you assessed developability (aggregation, stability, viscosity)?	Not yet **0** · In silico **0.4** · Experimentally **1**	O	R	R	R
BI03	CMC	What is your cell line status?	Not started **0** · Research cell line **0.4** · Clonal, GMP-suitable line **1**	–	O	R	R
BI04	Science and evidence	Have you assessed immunogenicity risk?	Not yet **0** · In silico **0.5** · In-vitro assays **1**	O	R	R	R

Cell therapy (3 scored)

ID	Dimension	Question	Answer options and points	D1	D2	D3	D4
CT01	–	What is your cell source?	Autologous · Allogeneic · iPSC-derived – not scored	R	R	R	R
CT02	CMC	What is your manufacturing process status?	Research process **0.3** · Closed or automated process in development **0.6** · GMP-compliant process locked **1**	O	R	R	R

ID	Dimension	Question	Answer options and points	D1	D2	D3	D4
CT03	CMC	Do you have a potency assay?	No **0** · Candidate assay **0.5** · Qualified assay **1**	O	R	R	R
CT04	Market	Have you defined vein-to-vein time and COGS targets?	No **0** · Yes **1**	–	O	R	R

Gene therapy and editing (3 scored)

ID	Dimension	Question	Answer options and points	D1	D2	D3	D4
GT01	–	How do you deliver your therapy?	AAV · Lentiviral · Lipid nanoparticle · Other non-viral – not scored	R	R	R	R
GT02	CMC	How is vector or delivery supply secured?	Research grade only **0.3** · CDMO capacity reserved **0.7** · GMP supply secured **1**	O	R	R	R
GT03	Science and evidence	Have you addressed pre-existing immunity and redosing?	No **0** · Plan **0.5** · Data **1**	O	R	R	R
GT04	Science and evidence	Have you assessed off-target or integration safety?	Not yet **0** · In silico **0.4** · Experimentally **1**	O	R	R	R

RNA and oligonucleotides (3 scored)

ID	Dimension	Question	Answer options and points	D1	D2	D3	D4
RN01	Science and evidence	How do you deliver to the target tissue?	No approach yet **0** · Novel delivery, no in-vivo data **0.3** · Novel delivery with in-vivo data **0.8** · Clinically validated delivery (e.g. GalNAc to liver) **1**	R	R	R	R
RN02	Product and plan	Is the duration of effect established in vivo?	No **0** · Yes **1**	O	R	R	R
RN03	Science and evidence	Is your chemistry optimised for tolerability?	No **0** · In progress **0.5** · Done **1**	O	R	R	R

Platform (3 scored)

ID	Dimension	Question	Answer options and points	D1	D2	D3	D4
PL01	Product and plan	How many programs has your platform generated?	One **0.4** · Two to three **0.7** · Four or more **1**	R	R	R	R
PL02	External validation	Has an external partner validated the platform?	No **0** · Research collaboration **0.6** · Paid deal **1** (public link)	O	O	O	O
PL03	–	What is your business model?	Own pipeline · Partnering · Hybrid – not scored	R	R	R	R
PL04	Product and plan	Have you nominated a lead asset?	No **0.5** · Yes **1**	R	R	R	R